

- Reach a nearby DNS server.
- Lesser network latency
- Better load distribution
- Improved availability.
- Automatic routing around failed locations.

1) TCP uses a receiver window to control how much data the sender can transmit before receiving acknowledgement.

Improvement:

- Increase TCP receive & send buffers appropriately.
- monitor window utilization
- Ensure sufficient memory and network capacity.
- Avoid unnecessary buffer exhaustion

2) Nagle's algorithm attempts to reduce network overhead by combining small TCP packets into larger packets. This is useful for normal applications because it reduces the number of packets.

Smaller order → TCP waits → packet aggregation  
 → Additional delay → Late acknowledgment

This allows small packets to be transmitted to immediately.

3) When a client establishes a TCP connection, the TCP three-way handshake.

SYN → SYN-ACK → ACK

If the server's SYN queue is too small, incoming connection requests may be dropped or delayed.

Encrypt  
audio / video  
compressate for  
traffic

#### DNS caching

Frequently requested domain records should be cached at recursive DNS resolvers and local DNS servers.

#### DNS Pre-fetching

Frequently accessed DNS records can be retrieved before their TTL expires, reducing the chance that users encounter cache misses.

### 3) Advantages and limitations

#### Advantages:

- Lower DNS latency
- Better performance for global users
- High availability
- Automatic traffic distribution

#### Limitations:

- Anycast infrastructure is more complex
- Requires multiple DNS locations
- Global DNS deployment can increase operational costs

3. A

#### 1) Impact of centralized DNS architecture

This creates a single point of failure and performance bottlenecks.

- i) DNS server overload.
- ii) Increased querying time
- iii) Poor user experience
- iv) Possible DNS service failure.

### Case Study

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1) The problem is primarily related to the transport protocol choice and application-level buffering, with the 8% packet loss and 180 ms jitter being major indicators.

Video conferencing is real-time multimedia application. It requires packets to arrive quickly and continuously. If TCP is used, lost packets are retransmitted. Although this provides reliability, retransmission in frozen video and delayed audio.

The 180 ms jitter indicates that packets are arriving at irregular intervals. While in sufficient buffering can cause audio/video interruptions.

2) Real-time applications such as video multimedia requirements.

- Low latency
- Low jitter
- Continuous data delivery
- Timely packet arrival
- Reasonable packet loss tolerance

Therefore, UDP is more suitable than TCP for the media stream. Modern real-time communication system commonly uses UDP-based mechanisms with application level techniques.



**SIMATS ENGINEERING  
ASSESSMENT TOOL 2  
Case study**

**COURSE NAME:** Computer Networks

**COURSE CODE:** CSA07

**COURSE OUTCOME COVERED**

**CO3:** Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions

Assessment Tool Weightage: 40%

**Code of Conduct:**

I T. Subahan(19251112) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

**Signature of the student:**

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